LOT-P1 ⬜⬜ 00-PLAN

00 — PLAN

Concept

A silent object on a desk. It receives **autonomous, AI-written notifications** from lot-systems.com (“Coffee time!”), shows them on a tiny screen, and returns one signal: press the button (“Copy”) and the site’s **Log tab** records it. It also measures weather/air and reports it. Two faces:

- **Face A — mirror:** polished stainless steel, nothing else.
- **Face B — interface:** camera, screen, button (behind a non-metal window; see conflict C1).

Form: flat silver square **40 × 40 × 5 mm**, two stainless parts (back plate + front frame), wireless charging.

Requirement traceability (your 19 points)

#	Requirement	Where addressed	Status
1	PCBWay	05 — PCB, SMT assembly, CNC steel all quotable at PCBWay	Planned
2	Pager-like notification from AI site	04 — scheduled job → Claude writes text → device poll/push	Designed
3	2-part stainless body	02 — 316L back plate + frame	Designed
4	Flat silver square 4×4 cm × 5 mm	02 — stack budget	**Tight, feasible**
5	Camera	01/03 — Himax HM0360-class ultra-thin module	Selected
6	LOT API connector	04 — `/api/device/*`	Designed (not built)
7	PDF manuals	`pdf/`, `build-pdf.sh`	Done (this session)
8	Compress info each session	`SESSION-DIGEST.md`	Done
9	Firmware documents	03	Done
10	Software to connect with firmware	04 (provisioning CLI + web pairing)	Designed
11	Separate documents	one file per topic	Done
12	Charger	01 — Qi puck, 5 V/1 A	Selected
13	100-unit run	05	Planned
14	Weather sensor	01 — SHT45 (T/RH) + BMP390 (pressure)	Selected
15	AI-grade off-the-shelf sensors	01 — BME688 (AI gas), BHI260AP (on-chip AI IMU)	Selected
16	“Copy” button → Log tab	04 — `POST /api/device/event {type:"copy"}`	Designed
17	One side polished steel	02 — #8 mirror, ~Ra 0.05 µm	Designed
18	Other side camera+screen+button	02	Designed
19	Wireless charging	02/01 — coil behind glass window	**Conflict C1**

Engineering verdict & conflicts (read these)

- **C1 — Steel blocks Qi and Wi-Fi.** A fully steel shell cannot be wirelessly charged or have a working antenna. Resolution: Face B carries a **glass/sapphire-ceramic window** covering screen + coil + antenna keep-out; charger sits on Face B. Face A stays pure polished steel. Alternative: ceramic (zirconia) back = no metal on Face B at all.
- **C2 — 5 mm is extremely thin** with camera + screen + battery + Wi-Fi MCU. Budget closes only at ~3.4 mm internal height with a ~90–120 mAh battery (≈ days of standby, not weeks, with camera idle). Camera is event-triggered only.
- **C3 — Privacy:** a camera on a notification object needs a hardware shutter/LED or “off unless asked” policy. Default: camera disabled in firmware, enabled per-user opt-in.
- **C4 — Radio compliance:** a 100-unit run with Wi-Fi/Qi needs FCC/CE pre-scan; use a **pre-certified module** (ESP32-C3-MINI-1).
- **C5 — Cameras at “AI grade”:** 320×320 low-power sensor supports presence/QR/gesture, not photography. Set expectations.

Design principles (brand-neutral until brand.lot-systems.com is read)

Monolithic, quiet, one gesture, one signal back. No logo on Face A except optional laser micro-mark on edge.